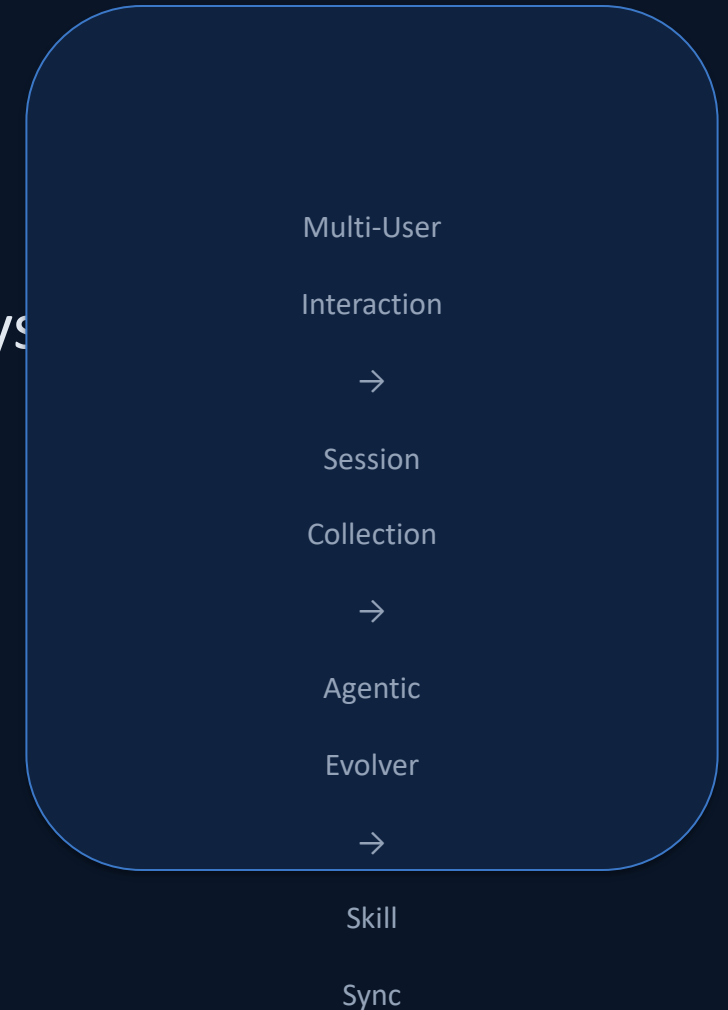


SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

arxiv.org/abs/2604.08377

ML Researchers & AI Systems Engineers



The Problem: Static Skills After Deployment

Current LLM agents rely on reusable skills manually installed from a centralized hub.

These skills never evolve from runtime experience — solutions are rediscovered per user, per session.

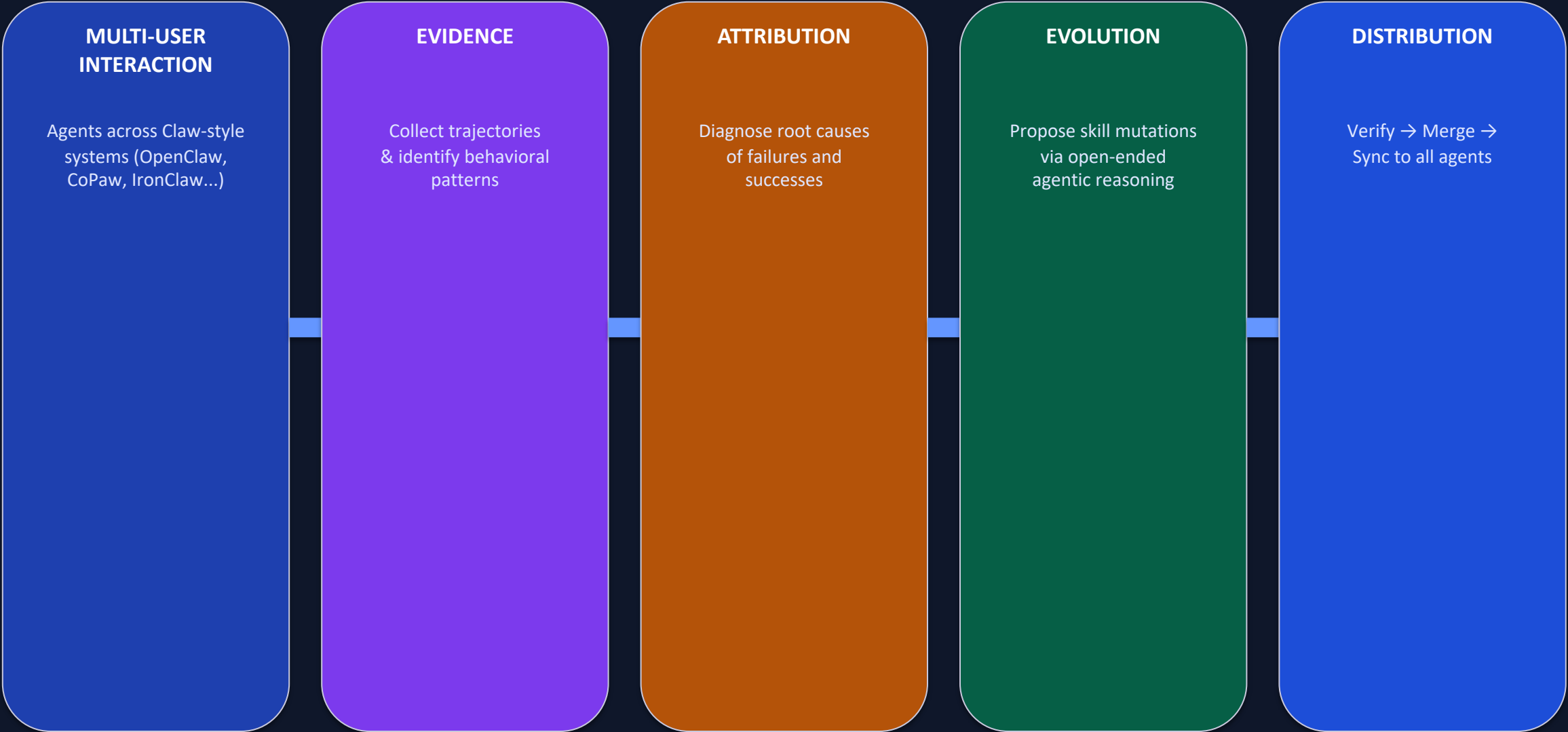
STATIC PARADIGM

- ✗ Skills fixed at deployment
- ✗ Failures repeat across users
- ✗ Solutions lost after each session
- ✗ Memory tied to specific instances
- ✗ Library is a static resource

SKILLCLAW EVOLVING PARADIGM

- ✓ Skills evolve from runtime
- ✓ Failures inform system-wide fixes
- ✓ Solutions accumulate across sessions
- ✓ Cross-user trajectory aggregation
- ✓ Shared, continuously improving pool

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Feedback Loop

Agentic Evolver: Evidence → Attribution → Evolution

Three-stage agentic workflow driven by open-ended reasoning, not predefined rules

1 EVIDENCE

Analyze trajectories across users and contexts

- Aggregate session trajectories grouped by skill
- Identify behavioral patterns (success & failure modes)
- Map conditions where skill works vs. breaks

2 ATTRIBUTION

Diagnose root causes of failures and successes

- Trace action–feedback causal chains
- Attribute outcomes to specific skill deficiencies
- Determine if improvement is generalizable or idiosyncratic

3 EVOLUTION

Propose skill mutations through agentic reasoning

- Refine existing skills or create new ones
- Produce updates via open-ended reasoning
- Pass candidates to validator before merge

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Different users exercising the same skill under diverse contexts produce complementary views of its behavioral boundary — conditions under which it works and fails.

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

Compatible Claw-style Systems:

OpenClaw

CoPaw

IronClaw

PicoClaw

ZeroClaw

NanoClaw

NemoClaw

Key Insight: Signal Diversity from Multi-User Aggregation

- Aggregated evidence reveals patterns invisible within a single user's session
- Complementary success/failure signals sharpen the evolver's attribution accuracy
- Generalizable improvements are separated from user-specific workarounds

Case Study: Multimodal Creative Task Pipeline

Accept

Reject

6 evolution attempts over 6 days — only 1 accepted (Conservative Verification)

Day	Candidate Skill	Skill Function / Change	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	Reject	Not admitted to next cycle

Result: 1 of 6 candidates accepted | Conservative verification prevents regressions

Case Study: Git Push Auth-Fallback Skill

Accept

Reject

Iterative refinement over 6 days — 3 accepted updates before reaching a plateau

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added 'push hang treated as auth failure'; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

Result: 3 of 6 days yielded accepted updates | Successful iterative refinement before plateau

Key Takeaways: Three Properties That Distinguish SkillClaw

1 COLLECTIVE EVOLUTION

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. Failures and solutions observed by any single user inform updates available to all agents.

2 FULLY AUTOMATIC

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. The agentic evolver autonomously processes evidence and generates updates.

3 AGENTIC EVOLUTION PARADIGM

Skill updates are produced through open-ended reasoning rather than predefined update rules. The evolver uses an autonomous agent to perform context-aware skill mutations.

Implications for Continuously Improving Agent Systems

- SkillClaw shifts agent systems from static libraries to living, self-improving ecosystems
- Evaluation on WildClawBench (Qwen3-Max backbone) shows substantial task improvement
- Cross-user aggregation enables knowledge accumulation impossible at the single-user level